ALGEBRA II: MISCELLANEOUS PROBLEMS (1 – 40)

1. The average of the numbers 5, 9, 10, 13, and a equals 11. Find the value of a.
a. 16
b. 17
c. 18
d. 20

2. A shirt is on sale for \$34.40, which is 20% off the original price. Find the original price of the shirt.
a. \$36.40
b. \$41.28
c. \$43.00
d. \$54.40

3. How many quarts of water does one add to 6 quarts of a 60% alcohol solution to create a 40% alcohol solution?
a. 2 qt.
b. 3 qt.
c. 6 qt.
d. 9 qt.

4. Express $(7 + 3i) – (2 – 3i)$ in the form $a + bi$ where a and b are real numbers.
a. $5 – 6i$
b. $5 + 6i$
c. 9
d. $9 + 5i$

5. Find the value of k that completes the square for $x^2 + 10x + k$.
a. 5
b. 10
c. 25
d. 100

6. Find the solutions to the equation: $|2x – 14| = 6$.
a. $x = –4, 10$
b. $x = 4, 10$
c. $x = 11, 17$
d. $x = 12, 14$

7. Solve the inequality $7 < 4x + 3 \leq 21$, and express the solution as an interval.
a. (1, 4.5]
b . [1.75, 7)
c . (7/3, 17)
d. [4.5, 5.25]

8. Solve the inequality $\frac{-18}{6x-42} > 0$, and express the solution as an interval

a. (–3, ∞)
b. (7, ∞)
c. (–∞,–3)
d. (–∞, 7)

9. Solve the inequality $2|x - 10| < 3$, and express the solution as an interval.
a. (7, 8.5)
b. (8.5, 10)
c. (8.5, 11.5)
d. (11.5, 13)

10. An object is shot upward with an initial velocity of 240 feet per second so that its height *s* (in feet) above the ground after *t* seconds is given by $s(t) = -16t^2 + 240t$. For what values of *t* will the object be at least 416 feet above the ground?
a. [2, 5]
b. [2, 13]
c. [3, 5]
d. [3, 13]

11. Find the distance d(*A*, *B*) between the points *A*(−1, 0) and *B*(4, 3).
a. $3\sqrt{2}$
b. 5
c. $\sqrt{34}$
d. 8

12. Determine the point *A*(x, y) so that the points *A*(x, y), *B*(0, 3), *C*(1, 0), *D*(7, 2) will be the vertices of a parallelogram.
a. *A*(−6, 1)
b. *A*((3, 7)
c. *A*(5, 6)

d. $A(6, 5)$

13. Find the midpoint of the line segment from $A(-2, 9)$ to $B(4, 5)$.
a. $C(1, 7)$
b. $D(3, 7)$
c. $P(4, 9)$
d. $Q(5, 9)$

14. Find the point on the positive y-axis that is a distance 5 from the point $P(3, 4)$.
a. $A(0, 6)$
b. $B(0,8)$
c. $C(6,0)$
d. $D(8,0)$

15. Find the x-intercept and y-intercept of the equation $5x - 3y = 30$.
a. 6 and 10 respectively
b. 10 and 6 respectively
c. 6 and −10 respectively
d. 6 and 2 respectively

16. Give the equation for the circle with center $C(3, -2)$ and radius 4.
a. $x^2 + y^2 = 52$
b. $(x-3)^2 + (y-2)^2 = 16$
c. $(x+3)^2 + (y-2)^2 = 42$
d. $(x-3)^2 + (y+2)^2 = 16$

17. Give the center of the circle with equation $x^2 + 2x + y^2 - 10y + 22 = 0$.
a. $A(2, 4)$
b. $B(1, 5)$
c. $C(-1, 5)$
d. $D(-2, 4)$

18. Find an equation for the line with slope 1/2 and y-intercept 3.
a. $x/2 - y = 3$
b. $-x + 2y = 6$
c. $x + 2y = 6$
d. $2x - y = 3$

19. Find the slope of the line through the points $A(-1, 6)$ and $B(5, 2)$
a. −1
b. −2/3
c. 2/3
d. 1

20. Find an equation for the line with y-intercept 3 that is perpendicular to the line $3y = 2x - 4$.
a. $2y = 6 - 3x$
b. $2y = 3x + 6$
c. $3y = 9 - 2x$
d. $3y = 2x + 9$

21. Fahrenheit and Celsius temperatures are related by the equation $F = 9C + 32$, where F is the temperature in degrees Fahrenheit and C is the temperature on the Celsius scale. If the temperature is a balmy 77° Fahrenheit, what is the temperature on the Celsius scale?
a. 25°
b. 33.8°
c. 43°
d. 45°

22. If $f(x) = x^2 + 5$, find $f(a+h) - f(a)$.
a. $2ah + h^2 + 10$
b. $2ah + h^2 + 5$
c. $2ah + h^2$
d. h^2

23. From a square piece of cardboard with width x inches, a square of width $x - 3$ inches is removed from the center. Write the area of the remaining piece as a function of x.
a. $f(x) = 6x - 9$
b. $f(x) = 6x + 9$
c. $f(x) = 2x^2 - 9$
d. $f(x) = 2x^2 - 6x - 9$

24. If $P(4, -5)$ is a point on the graph of the function $y = f(x)$, find the corresponding point on the graph of $y = 2f(x - 6)$.
a. $A(1, 8)$
b. $B(2, -5)$
c. $C(6, 8)$
d. $D(10, -10)$

25. Explain how the graph of $y - 5 = (x - 3)^2$ can be obtained from the graph of $y = x^2$.
a. Shift the graph of $y = x^2$ left 3 units and down 5 units
b . Shift the graph of $y = x^2$ left 3 units and up 5 units
c . Shift the graph of $y = x^2$ right 3 units and down 5 units
d. Shift the graph of $y = x^2$ right 3 units and up 5 units

26. Determine e the vertex of $y = x^2 - 8x + 22$.
a. $A(-4, 11)$
b . $B(-4, 18)$
c . $C(4, 6)$
d . $D(4, 8)$

27. An object is projected upward from the top of a tower. Its distance in feet above the ground after t seconds is given by $s(t) = -16t^2 + 64t + 80$. How many seconds will it take to reach ground level?
a. 1 second
b . 4 seconds
c . 5 seconds
d . 8 seconds

28. Find the maximum value of $y = -x^2 + 6x$.
a. 8
b . 9
c . 10
d . 11

29. Several values of the two function *f* and *g* are listed in the following tables:

X	4 5 6 7
f(x)	7 6 5 4

X	4 5 6 7
g(x)	6 7 4 5

Find (*f* o *g*)(6).
a. 4
b . 5
c . 6
d . 7

30. Given $f(x) = 5x + 7$ and $g(x) = x^2 + 7$, find $(g \circ f)(x)$.
a. $(g \circ f)(x) = 5x^2 + 7$
b . $(g \circ f)(x) = 5x^2 + 42$
c . $(g \circ f)(x) = (5x)^2 + 14$
d . $(g \circ f)(x) = 25x^2 + 70x + 56$

31.
Find x:

$$\begin{bmatrix} 1 & 3x & 2 \\ 3 & 0 & 5 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 2 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 16 & 3 \\ 6 & 2 \end{bmatrix} = 3 \begin{bmatrix} x-2 & 0 \\ 2 & 2 \end{bmatrix}$$

a) 3
b) 2
c) 1
d) 1/4

32.

Which of the following is a factor of $2x^4 - 6x^3 - 20x^2 + 48x$

a) x - 3
b) x + 3
c) x – 1
d) x + 1

33.

Solve for x: $\frac{2x+3}{x-1} - \frac{2x-3}{x+1} = \frac{10}{x^2-1}$

a) No solution
b) x = 1
c) x = 2
d) x = ¾

34.

When photographing a lighted object, the exposure time (*t*) is inversely proportional to the square of the object's distance (*d*) from the light source. Which equation best represents this relationship?

a) $t^2 = kd$

b) $t = \frac{k}{d^2}$

c) $t = \frac{k}{d}$

d) $t = kd$

35.

Express as a single log $2\log_3 5 - \log_3 10 + 3\log_3 4$

a) $\log_3 \frac{5}{128}$
b) $\log_3 16$
c) $\log_3 160$
d) $\log_3 64$

36.

If $\log_7(x+1)+\log_7(x-5)=1$, then x =

a) -6
b) -2
c) 6
d) 7

37.

Find the fifth term in the expansion of $(x + y)^{10}$

a) $252x^5y^5$
b) $210x^4y^6$
c) $210x^6y^4$
d) $252x^6y^4$

38.

Solve the equation $3x^2 = 57x$ using the method of your choice. You might want to try graphing or the quadratic formula. The value(s) of x that solve the equation are:

a. $x = 19$ **b.** $x = -19$ **c.** $x = 19$ and $x = 0$ **d.** $x = -19$ and $x = 0$ **e.** $x = 0$

39.

Find $g(-1)$ if $g(x) = -\left(\frac{1}{2}\right)^x$.

a. $-\frac{1}{2}$ **b.** $\frac{1}{2}$ **c.** 0 **d.** -2 **e.** 2

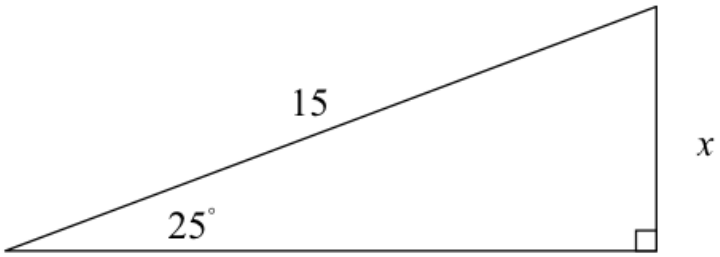
40.

Find the length of side x of the triangle below.

a. 13.6

b. 16.6

c. 6.3

d. 15.0

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